

MACHINE LEARNING PROJECT REPORT

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PROBLEM 1 : ELECTION DATA

You are hired by one of the leading news channels CNBE who wants to analyze recent elections. This survey was conducted on 1525 voters with 9 variables. You have to build a model, to predict which party a voter will vote for on the basis of the given information, to create an exit poll that will help in predicting overall win and seats covered by a particular party.

DATA ATTRIBUTES

1. vote: Party choice: Conservative or Labour
2. age: in years
3. economic.cond.national: Assessment of current national economic conditions, 1 to 5.
4. economic.cond.household: Assessment of current household economic conditions, 1 to 5.
5. Blair: Assessment of the Labour leader, 1 to 5.
6. Hague: Assessment of the Conservative leader, 1 to 5.
7. Europe: an 11-point scale that measures respondents' attitudes toward European integration. High scores represent 'Eurosceptic' sentiment.
8. political.knowledge: Knowledge of parties' positions on European integration, 0 to 3.
9. gender: female or male.

SAMPLE OF THE DATASET

Unnamed: 0	vote	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	gender
0	1 Labour	43	3	3	4	1	2	2	female
1	2 Labour	36	4	4	4	4	5	2	male
2	3 Labour	35	4	4	5	2	3	2	male
3	4 Labour	24	4	2	2	1	4	0	female
4	5 Labour	41	2	2	1	1	6	2	male

Table 1. sample of dataset (problem 1)

There are 10 variables in the dataset but “Unnamed” is not useful so we are dropping it thus, the dataset now has 9 variables in total.

1.1 Read the dataset. Do the descriptive statistics and do the null value condition check. Write an inference on it.

	vote	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	gender
0	Labour	43	3	3	4	1	2	2	female
1	Labour	36	4	4	4	4	5	2	male
2	Labour	35	4	4	5	2	3	2	male
3	Labour	24	4	2	2	1	4	0	female
4	Labour	41	2	2	1	1	6	2	male

Table 2. new sample of dataset

There are total 9 variables in this dataset now.

Check for the shape of the dataframe.

The dataset has 1525 rows and 9 columns after dropping Unnamed column.

Check for the data type.

vote	object
age	int64
economic.cond.national	int64
economic.cond.household	int64
Blair	int64
Hague	int64
Europe	int64
political.knowledge	int64
gender	object

Two variables are of object data type while 7 variables are of integer data type.

Check for the missing values in the dataset.

vote	1525 non-null
age	1525 non-null
economic.cond.national	1525 non-null
economic.cond.household	1525 non-null
Blair	1525 non-null
Hague	1525 non-null
Europe	1525 non-null
political.knowledge	1525 non-null
gender	1525 non-null

There are no missing values present in the dataset.

Check for duplicates

There are 8 duplicate records present in the dataset so we are dropping them. After dropping these duplicates the resulting shape is 1517 rows and 9 columns.

Summary stats

Descriptive statistics help describe and understand the features of a specific dataset by giving short summaries about the sample and measures of the data. The most recognized types of descriptive statistics are measures of centre : the mean, median and mode.

	count	mean	std	min	25%	50%	75%	max
age	1517.0	54.241266	15.701741	24.0	41.0	53.0	67.0	93.0
economic.cond.national	1517.0	3.245221	0.881792	1.0	3.0	3.0	4.0	5.0
economic.cond.household	1517.0	3.137772	0.931069	1.0	3.0	3.0	4.0	5.0
Blair	1517.0	3.335531	1.174772	1.0	2.0	4.0	4.0	5.0
Hague	1517.0	2.749506	1.232479	1.0	2.0	2.0	4.0	5.0
Europe	1517.0	6.740277	3.299043	1.0	4.0	6.0	10.0	11.0
political.knowledge	1517.0	1.540541	1.084417	0.0	0.0	2.0	2.0	3.0

Table 3. summary (problem 1)

From the above descriptive statistics we can conclude :

- Almost all variables are normally distributed as mean and median align in almost all the cases.
- Here only age is a continuous variable others are categorical in nature.
- The average rating for national economic conditions, household economic conditions and blair is 3.
- The average rating for knowledge of parties' position on Europe integration is 1.5

1.2 Perform Univariate and Bivariate Analysis. Do exploratory data analysis. Check for Outliers.

UNIVARIATE ANALYSIS :

Numerical data

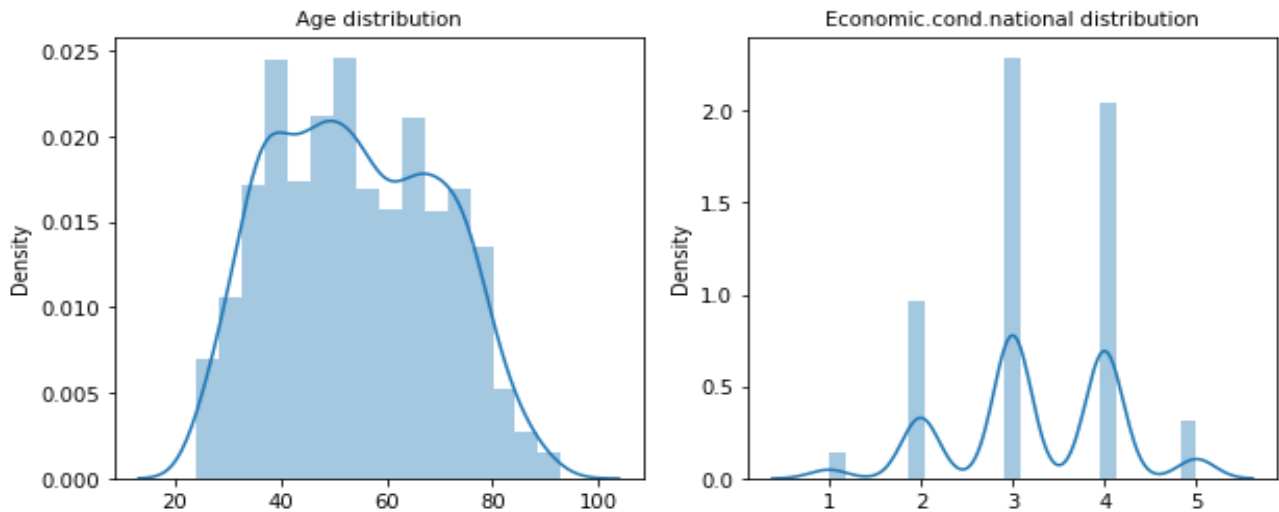


Fig 1. Univariate (age, eco.cond.national)

- The distribution of age is almost symmetrical thus following normal distribution.
- The distribution of Economic.cond.national is discrete as it is showing the ratings ranging from 1 to 5.

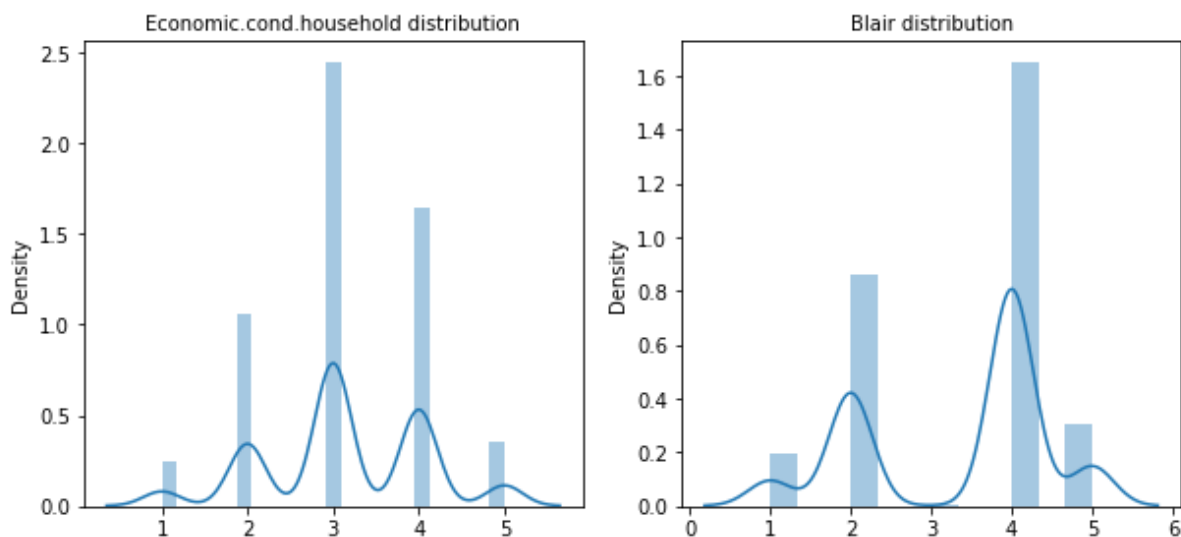


Fig 2. Univariate (eco.cond.household, blair)

- The distribution of Economic.cond.household is discrete as it is showing the ratings ranging from 1 to 5.
- The distribution of Blair is discrete as it is showing the ratings ranging from 1 to 5.

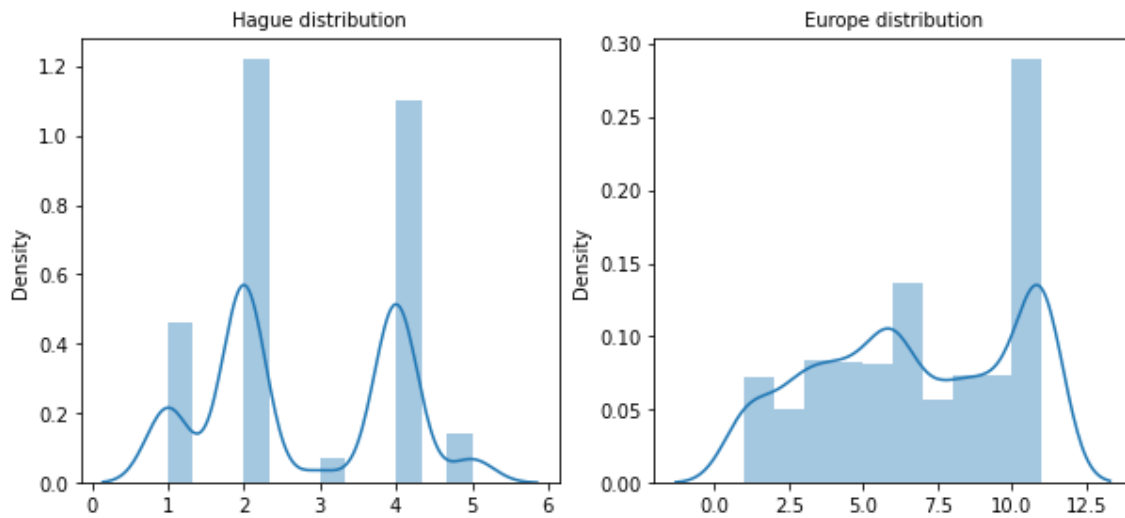


Fig 3. Univariate (hague, Europe)

- The distribution of Hague is discrete as it is showing the ratings.
- The distribution of Europe is slightly skewed towards right.

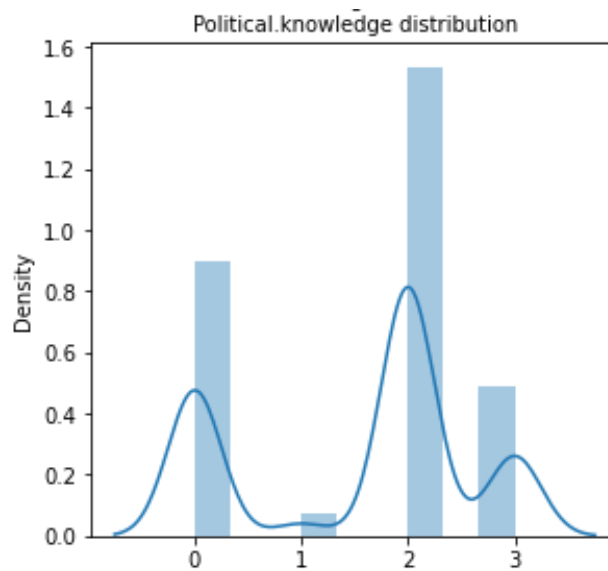


Fig 4. Univariate (political knowledge)

- The distribution of Political knowledge is discrete as it is showing the ratings.

Categorical data

The categorical data present in the above problem is gender and vote.

- Gender here is classified as male and female.
- Vote here is classified as labour and conservative.
- Countplot is used to represent the categorical data visually.

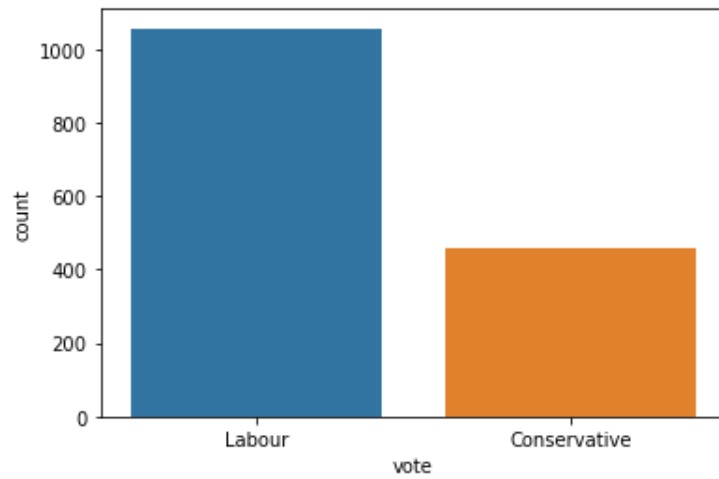


Fig 5. Countplot (vote)

- Around 1050 people vote for “Labour party” while around 460 vote for “Conservative party”

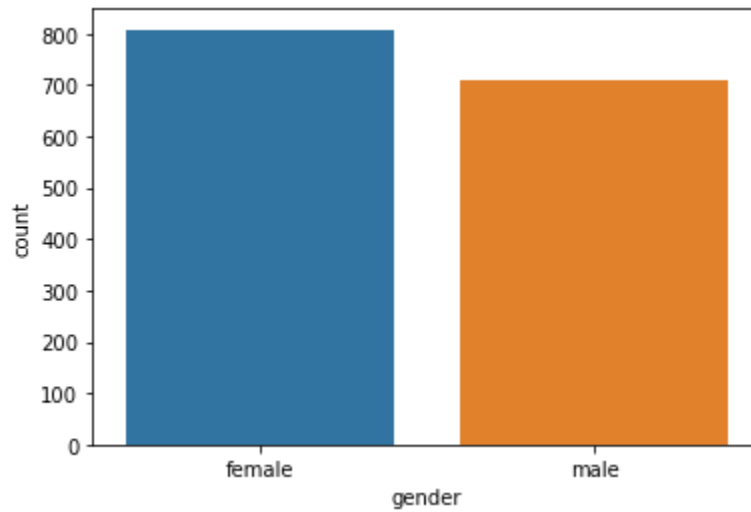


Fig 6. Countplot (gender)

- Around 800 female voters are there and 700 male voters are there.

BIVARIATE ANALYSIS

- In bivariate analysis all the numerical variables are compared to vote as the target variable here is “vote”
- We are using scatterplot to compare vote with respect to age.
- We will be using countplot with hue as “vote” for comparing other numerical columns.

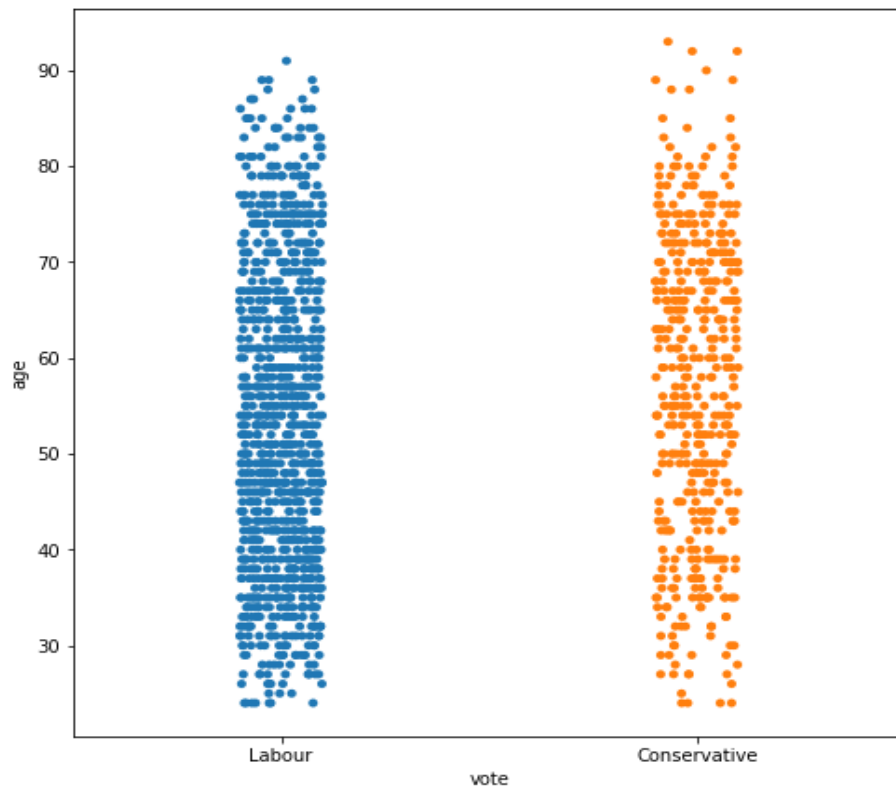


Fig 7. Scatterplot (age-vote)

- The above scatterplot depicts the relationship between age and vote.
- Those who are 90 and above vote for Conservative compared to Labour.
- More people vote for Labour as the data points are less scattered in Labour compared to Conservative.

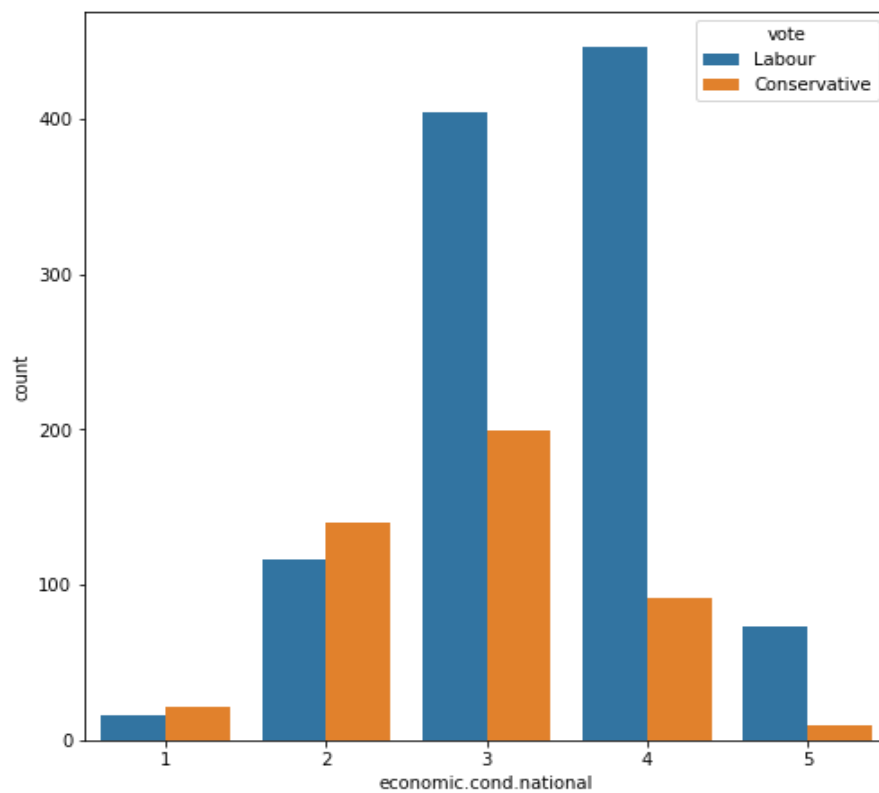


Fig 8. Countplot (eco.cond.national – vote)

- From the above countplot we can conclude that 1 rating is given to Conservative in current national economic conditions while Labour is given more ratings as 3,4 and 5.

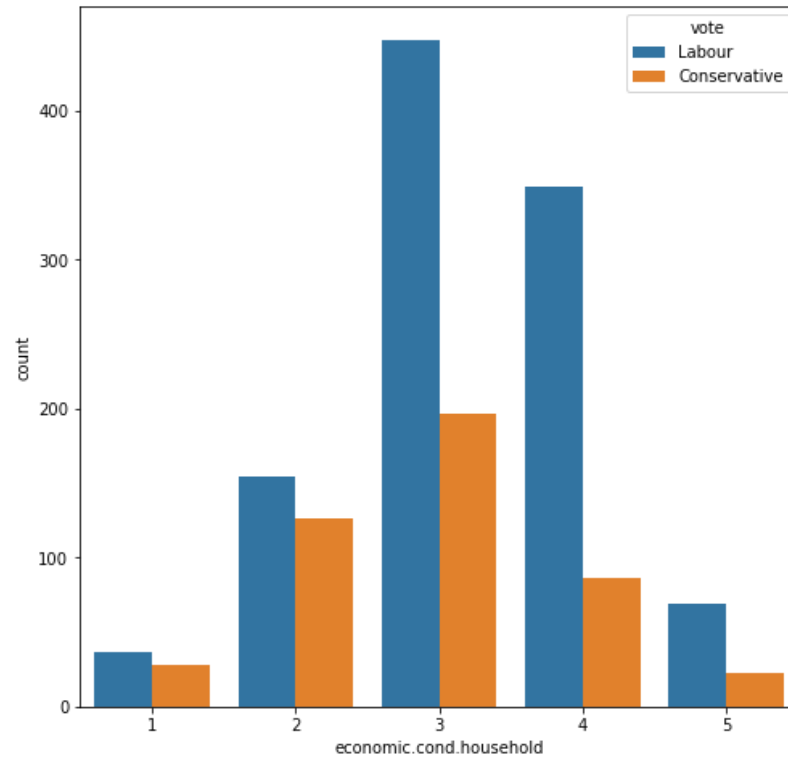


Fig 9. Countplot (eco.cond.household-vote)

- In assessment of current household economic conditions Labour has got highest rating as 5.
- The lowest rating i.e 1 is also given more to Labour party.

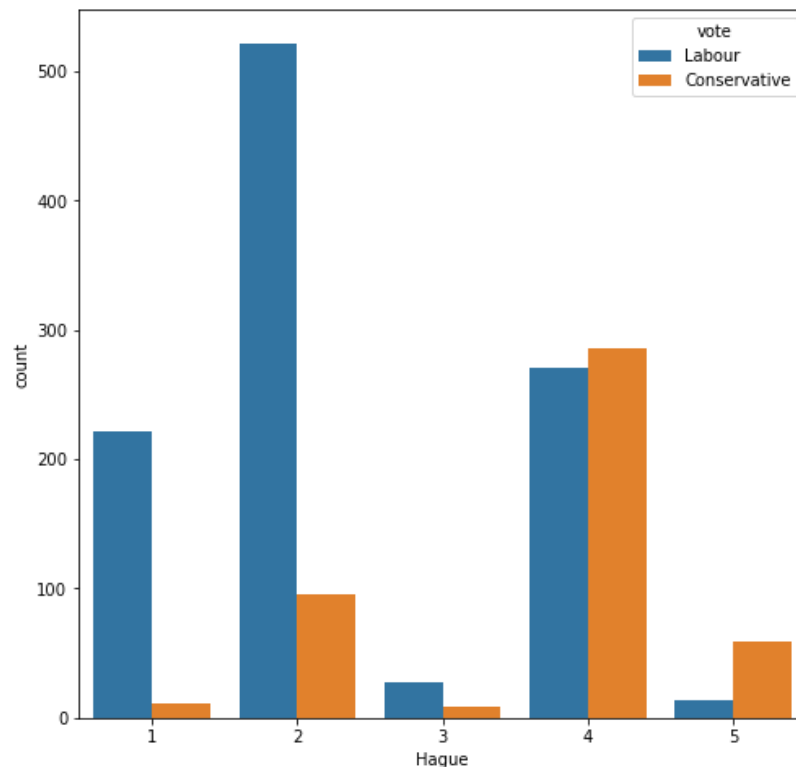


Fig 10. Countplot (hague-vote)

PAIRPLOT

Pairplot shows the relationship between the variables in the form of scatterplot and the distribution of the variable in the form of histogram or density function.

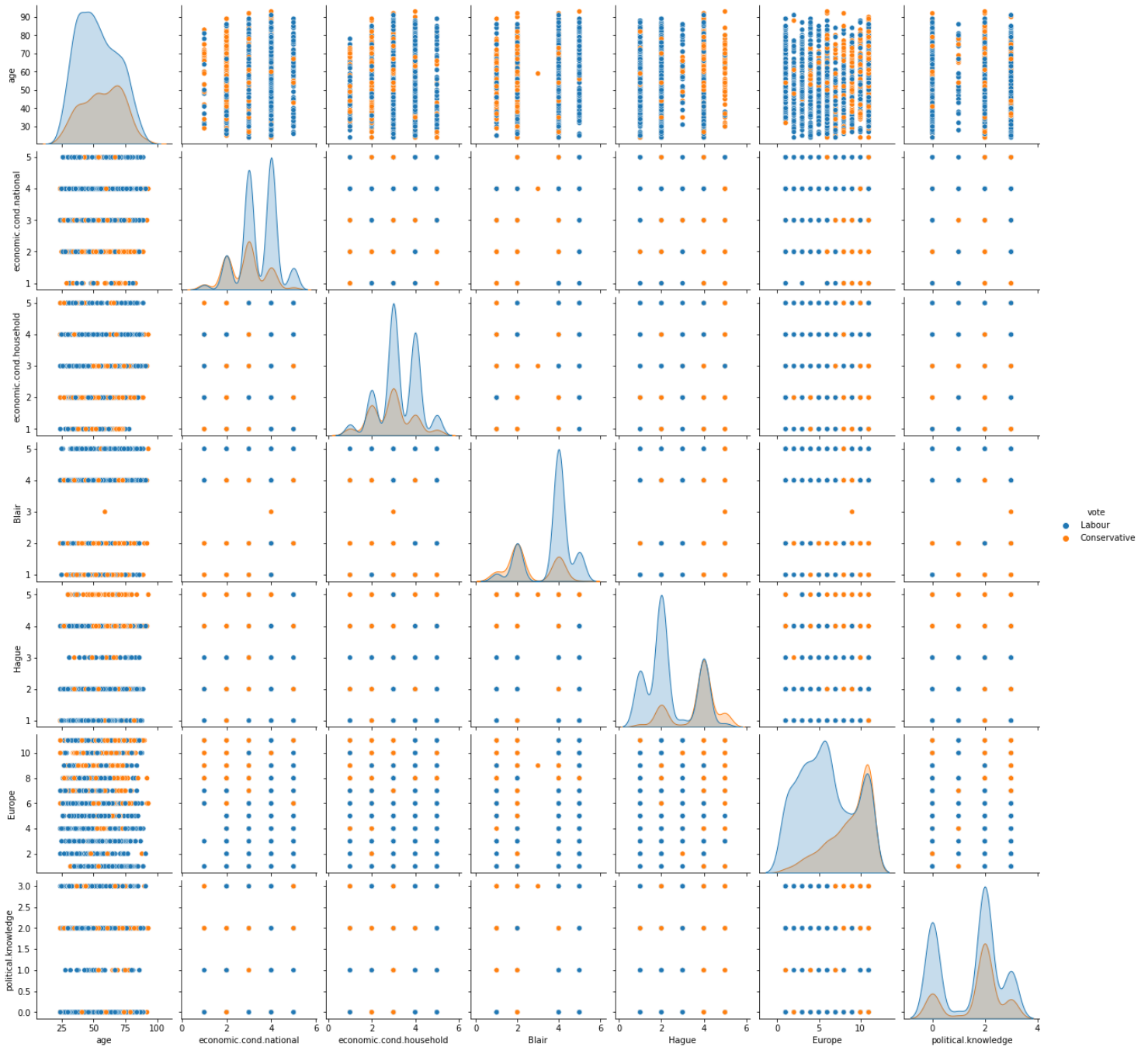


Fig 11. Pairplot

- From the above pairplot we can infer that there exists no correlation among the different variables,
- The distribution of age is approximately normal.
- Variables other than age are discrete in nature thus, the distribution is bimodal.
- The distribution of Europe is slightly skewed.

CORRELATION HEATMAP



Fig 12. Correlation heatmap

- From the above correlation heatmap it can also be concluded that there exists no correlation among the variables.

BOXPLOT

Boxplot displays the five-number summary of a dataset. The five number summary is the minimum, first quartile, median, third quartile and maximum. It also displays the outliers proportion in the dataset.

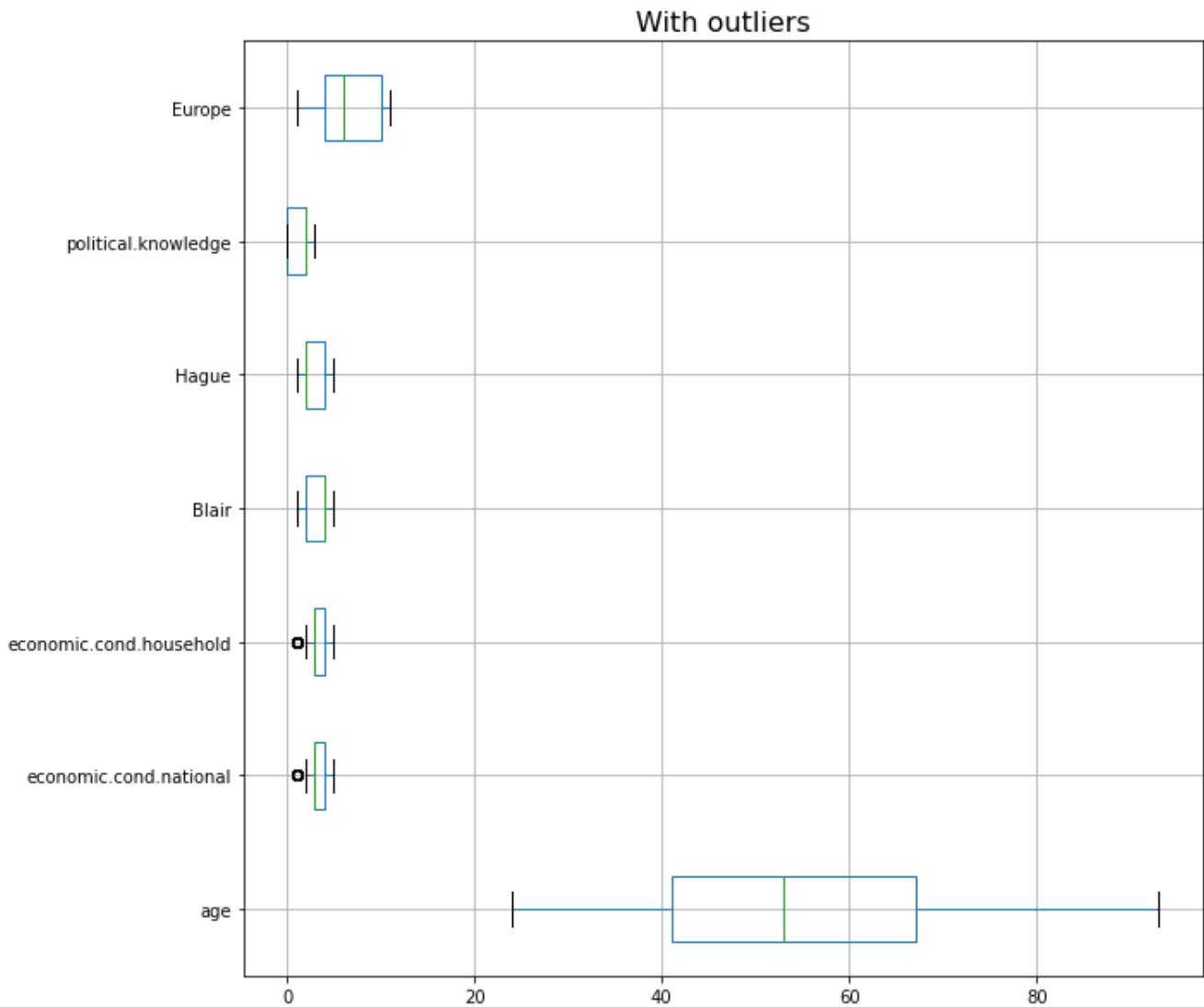


Fig 13. Boxplot

- From the above boxplot we conclude that no outliers are present in the dataset in almost all the variables.
- Economic.cond.household and economic.cond.national can have one outlier.
- Almost all the variables are symmetric but Europe is little skewed.

1.3 Encode the data (having string values) for Modelling. Is Scaling necessary here or not? Data Split: Split the data into train and test (70:30).

- We are **encoding** “gender” and “vote” columns.
 - ❖ For encoding “**gender**” **dummy** columns are created where **0** denotes **female** and **1** denotes **male**.
 - ❖ For encoding “**vote**” **label encoding** is used where **0** denotes **conservative** and **1** denotes **labour**.

	vote	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	gender_male
0	1	43	3	3	4	1	2	2	0
1	1	36	4	4	4	4	5	2	1
2	1	35	4	4	5	2	3	2	1
3	1	24	4	2	2	1	4	0	0
4	1	41	2	2	1	1	6	2	1

Table 4. encoded dataset sample

- **Scaling**

- ❖ Scaling is necessary here only for **LDA** (Linear Discriminant Analysis) and **KNN** rest other models are built without scaling for example, Logistic Regression, Naïve Bayes.
- ❖ The column age has two-digit numbers while other variables are in single digit and they almost range between 0-11 but the column age has 93 as its maximum value thus, scaling is necessary here.
- ❖ The standard deviation of age is around 15.7 but rest of the variables has standard deviation between 0 to 3 thus making scaling even more appropriate for distance based model like KNN.

- **Data split**

- ❖ The data is splitted in 70 : 30 ratio for the given problem.
- ❖ Random state for train_test_split is taken as 1.

1.4 Apply Logistic Regression and LDA (linear discriminant analysis).

Logistic Regression model :

- Here vote is the target variable.
- The model is build without treating the outliers and without scaling.
- The attributes taken for building the model are default values.

Model performance (training set) :

- ❖ Training set accuracy : 0.836
- ❖ Auc_roc score : 0.890

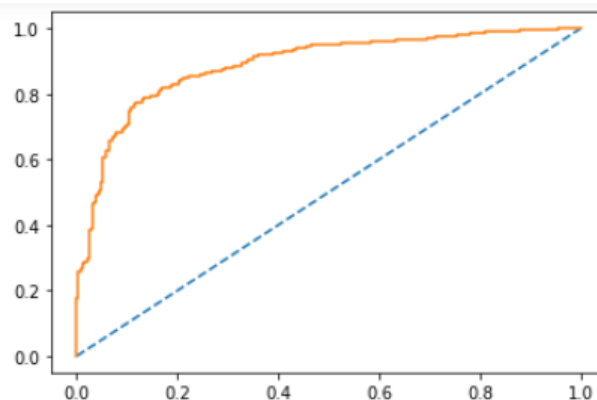


Fig 14. AUC_roc curve logistic reg training

	precision	recall	f1-score	support
0	0.75	0.65	0.70	307
1	0.86	0.91	0.89	754
accuracy			0.84	1061
macro avg	0.81	0.78	0.79	1061
weighted avg	0.83	0.84	0.83	1061

Table 5. classification report (LR-training)

- ❖ The recall for training set of labour party (1) is 0.91 while that of conservative (0) is 0.65.
- ❖ The precision of labour party is 0.86 while that of conservative is 0.75

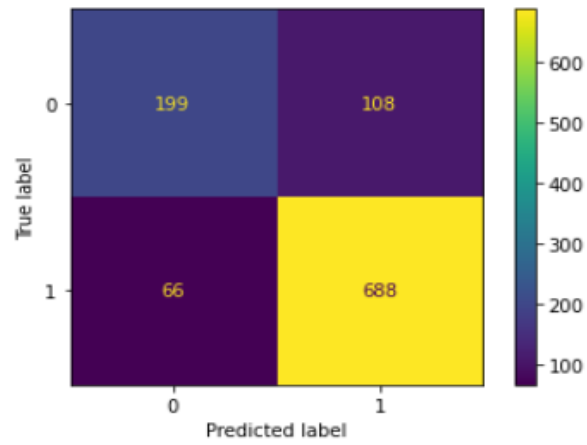


Fig 15. Confusion matrix (LR-training)

- ❖ The value of false positives is 66 while that of false negatives is 108.

Model performance (testing set) :

- ❖ Testing set accuracy : 0.826
- ❖ Auc_roc score : 0.877

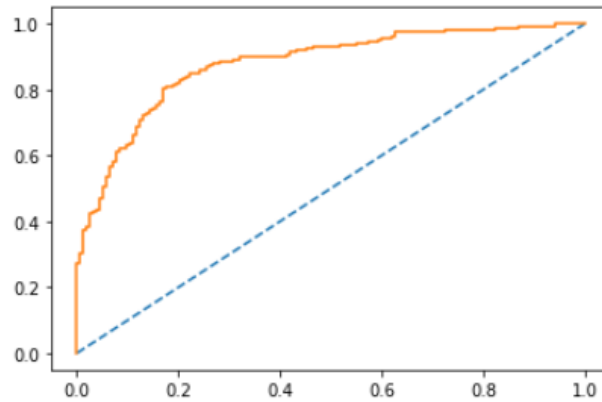


Fig 16. Auc_roc curve (LR-testing)

	precision	recall	f1-score	support
0	0.75	0.72	0.74	153
1	0.86	0.88	0.87	303
accuracy			0.83	456
macro avg	0.81	0.80	0.80	456
weighted avg	0.83	0.83	0.83	456

Table 6. classification report (LR-testing)

- ❖ The recall of labour party (1) is 0.88 while that of conservative (0) is 0.72
- ❖ The precision of the labour party is 0.86 while that of conservative is 0.75

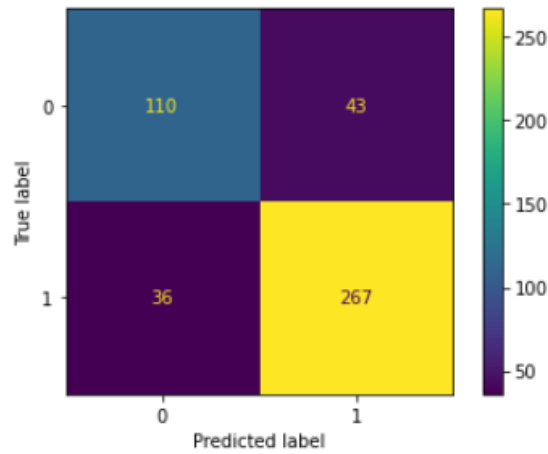


Fig 17. Confusion matrix (LR-testing)

- ❖ The value of false positives in the testing dataset is 36 while that of false negatives is 43.

CONCLUSION :

- ❖ The model is fine as there is no under fitting or over fitting in the basic model of logistic regression.
- ❖ Here recall and precision both are important.
- ❖ The recall of training set for labour (1) is 0.91 while that of testing is 0.88 thus, the model is working fine.
- ❖ The precision of training set for labour (1) is 0.86 while that of testing is 0.86 too.

LDA (linear discriminant analysis):

- Vote is the target variable.
- The model is build with outliers and “age” column is also scaled using zscore method.
- The attributes taken are default values.

Model performance (training set) :

- ❖ Training set accuracy : 0.834
- ❖ Auc_roc score : 0.889

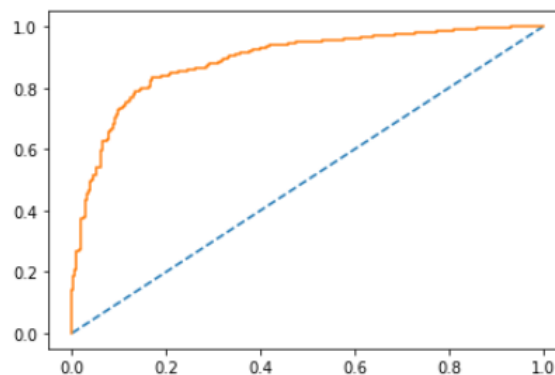


Fig 18. Auc_roc curve (LDA-training)

	precision	recall	f1-score	support
0	0.74	0.65	0.69	307
1	0.86	0.91	0.89	754
accuracy			0.83	1061
macro avg	0.80	0.78	0.79	1061
weighted avg	0.83	0.83	0.83	1061

Table 7. classification report (LDA-training)

- ❖ The recall for labour party (1) is 0.91 while that of conservative party (0) is 0.65
- ❖ The precision for labour party is 0.86 while that of conservative party is 0.74

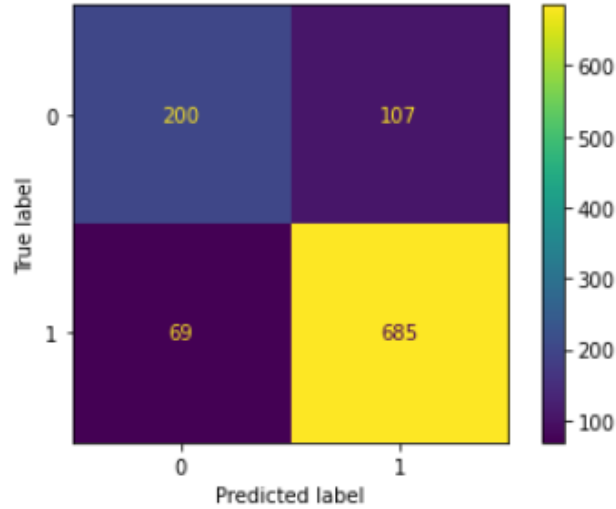


Fig 19. Confusion matrix (LDA-training)

- ❖ The value of false positives is 69 and that of false negatives is 107

Model performance (testing set) :

- ❖ Testing set accuracy : 0.833
- ❖ Auc_roc score : 0.888

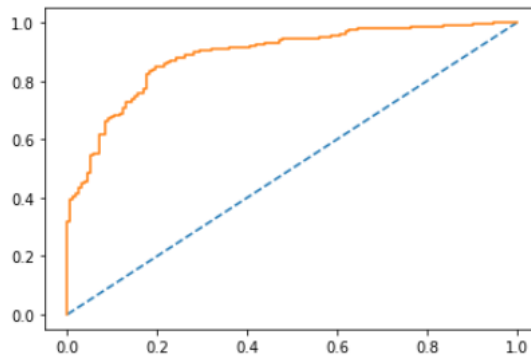


Fig 20. Auc_roc curve (LDA-testing)

	precision	recall	f1-score	support
0	0.77	0.73	0.74	153
1	0.86	0.89	0.88	303
accuracy			0.83	456
macro avg	0.82	0.81	0.81	456
weighted avg	0.83	0.83	0.83	456

Table 8. classification report (LDA-testing)

- ❖ The recall of labour party (1) is 0.86 while that of conservative is 0.73
- ❖ The precision of labour party is 0.86 while that of conservative is 0.77

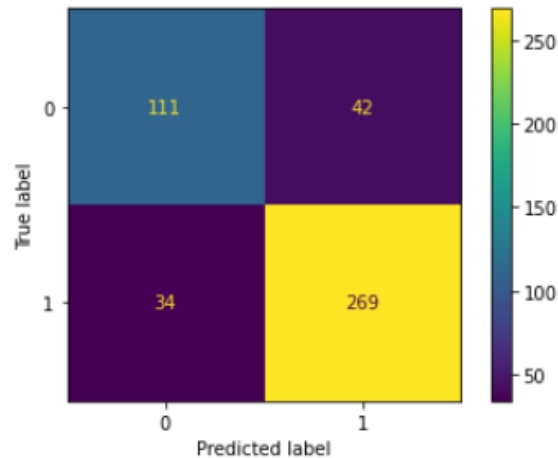


Fig 21. Confusion matrix (LDA-testing)

- ❖ The value of false positives is 34 while that of false negatives is 42.

CONCLUSION :

- ❖ The model is fine as there is no under fitting or over fitting in the basic model of LDA.
- ❖ Here recall and precision both are important.
- ❖ The recall of training set for labour (1) is 0.91 while that of testing is 0.89 thus, the model is working fine.
- ❖ The precision of training set for labour (1) is 0.86 while that of testing is 0.86 too.

1.5 Apply KNN Model and Naïve Bayes Model. Interpret the results.

Naïve Bayes :

- Here vote is the target variable.
- The model is build without treating the outliers and without scaling.
- The attributes taken for building the model are default values.

Model performance (training set) :

- ❖ Training set accuracy : 0.835
- ❖ Auc_roc score : 0.888

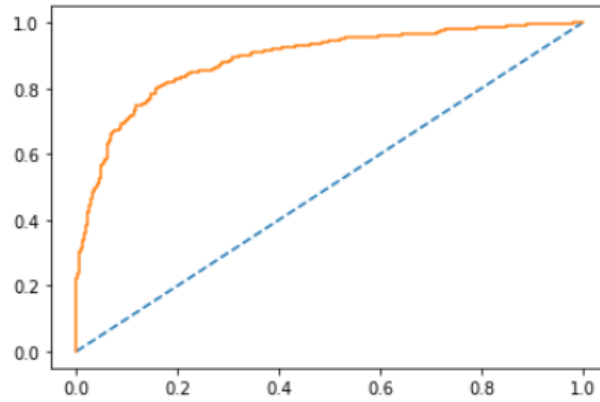


Fig 22. Auc_roc(Naïve bayes-training)

	precision	recall	f1-score	support
0	0.73	0.69	0.71	307
1	0.88	0.90	0.89	754
accuracy			0.84	1061
macro avg	0.80	0.79	0.80	1061
weighted avg	0.83	0.84	0.83	1061

Table 9. classification report (naïve bayes-training)

- ❖ The recall for labour party (1) is 0.90 while that of conservative party is 0.88
- ❖ The precision for labour party is 0.88 while that of conservative is 0.73

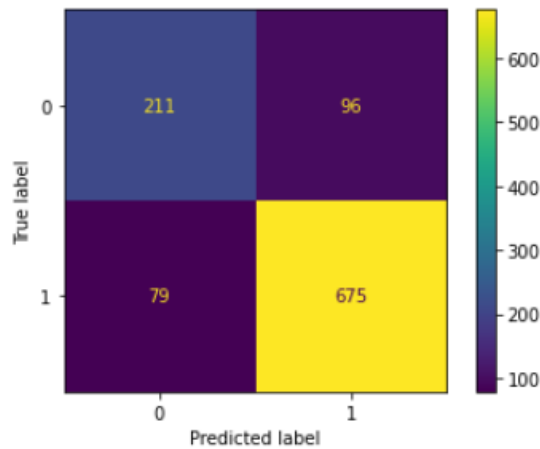


Fig 23. Confusion matrix (naïve bayes-training)

- ❖ The value of false positives is 79 while that of false negatives is 96

Model performance (testing set) :

- ❖ Testing set accuracy : 0.822
- ❖ Auc_roc score : 0.876

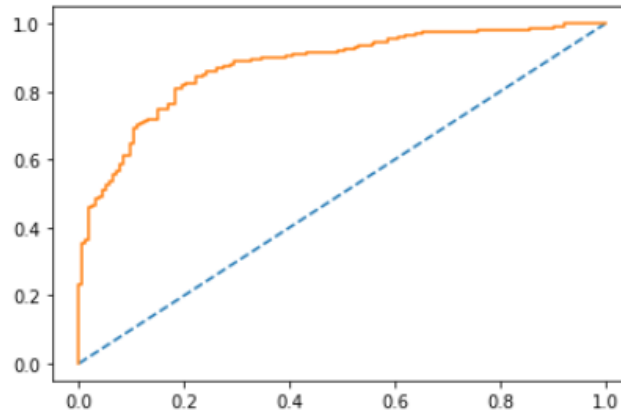


fig 24. Auc_roc curve (naïve bayes-testing)

	precision	recall	f1-score	support
0	0.74	0.73	0.73	153
1	0.87	0.87	0.87	303
accuracy			0.82	456
macro avg	0.80	0.80	0.80	456
weighted avg	0.82	0.82	0.82	456

Table 10. classification report (naïve bayes-testing)

- ❖ The recall for labour party (1) is 0.87 while that of conservative is 0.73
- ❖ The precision for labour party (1) is 0.87 while that of conservative is 0.74

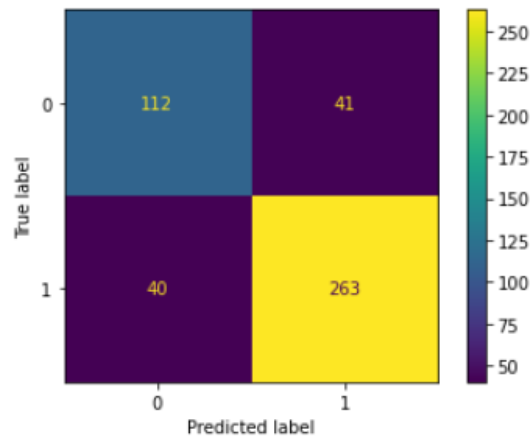


Fig 25. Confusion matrix (naïve bayes-testing)

- ❖ The value of false positives is 40 while that of false negatives is 41

CONCLUSION :

- ❖ The model is fine as there is no under fitting or over fitting in this model.
- ❖ Here recall and precision both are important.
- ❖ The recall of training set for labour (1) is 0.90 while that of testing is 0.87 thus, the model is working fine.
- ❖ The precision of training set for labour (1) is 0.88 while that of testing is 0.87 too.

KNN (K-nearest neighbor) :

- Here vote is the target variable.
- The model is build without treating the outliers and with scaling using zscore.
- The attributes taken for building the model are default values.

age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	gender_male
-0.716161	-0.278185	-0.148020	0.565802	-1.419969	-1.437338	0.423832	-0.936736
-1.162118	0.856242	0.926367	0.565802	1.014951	-0.527684	0.423832	1.067536
-1.225827	0.856242	0.926367	1.417312	-0.608329	-1.134120	0.423832	1.067536
-1.926617	0.856242	-1.222408	-1.137217	-1.419969	-0.830902	-1.421084	-0.936736
-0.843577	-1.412613	-1.222408	-1.988727	-1.419969	-0.224465	0.423832	1.067536

Table 11. scaled dataset

Model performance (training set) :

- ❖ Training set accuracy : 0.855
- ❖ Auc_roc score : 0.90

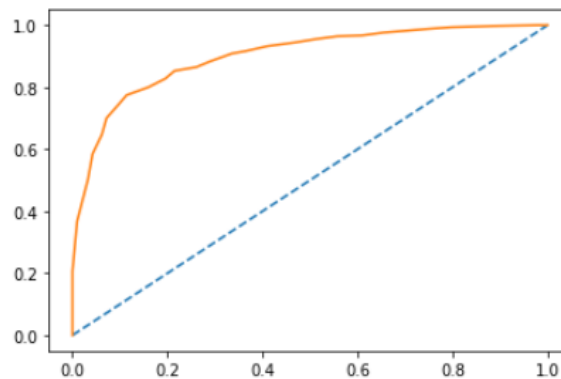


Fig 26. Auc_roc curve(KNN-training)

	precision	recall	f1-score	support
0	0.77	0.71	0.74	307
1	0.89	0.92	0.90	754
accuracy			0.86	1061
macro avg	0.83	0.81	0.82	1061
weighted avg	0.85	0.86	0.85	1061

Table 12. classification report (KNN-training)

- ❖ The recall for labour is 0.92 while that of conservative is 0.71
- ❖ The precision of labour is 0.89 while that of conservative is 0.77

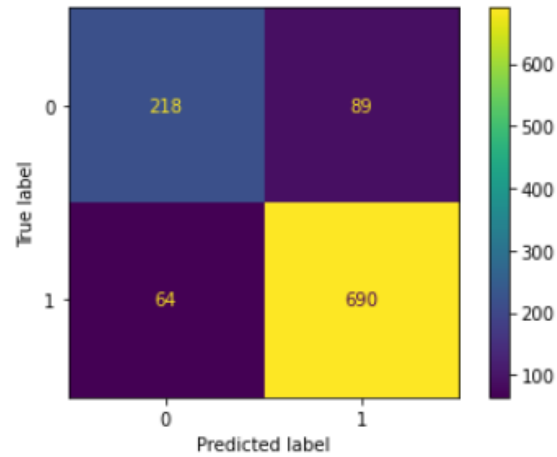


Fig 27. Confusion matrix (KNN-training)

- ❖ The value of false positives is 64 while that of false negatives is 89

Model performance (testing set) :

- ❖ Testing set accuracy : 0.824
- ❖ Auc_roc score : 0.887

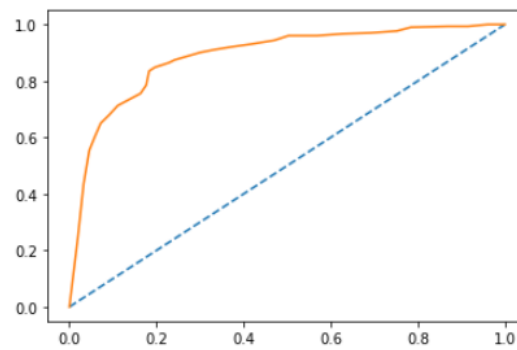


Fig 28. Auc_roc curve (KNN-testing)

	precision	recall	f1-score	support
0	0.77	0.69	0.72	153
1	0.85	0.89	0.87	303
accuracy			0.82	456
macro avg	0.81	0.79	0.80	456
weighted avg	0.82	0.82	0.82	456

Table 13. classification report (knn-testing)

- ❖ The recall for labour is 0.89 while that of conservative is 0.69
- ❖ The precision for labour is 0.85 while that of conservative is 0.77

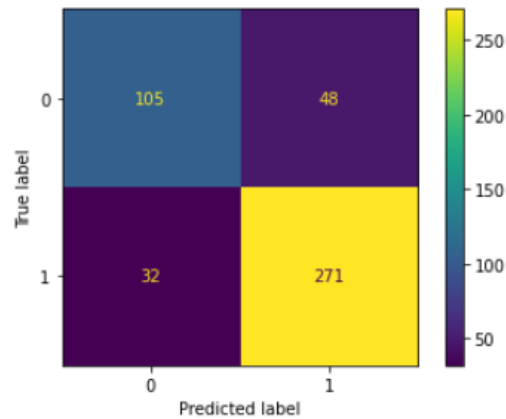


Fig 29. Confusion matrix (knn-testing)

- ❖ The value of false positives is 32 while that of false negatives is 48

CONCLUSION :

- ❖ The model is fine as there is no under fitting or over fitting in the basic model of KNN.
- ❖ Here recall and precision both are important.
- ❖ The recall of training set for labour (1) is 0.92 while that of testing is 0.89 thus, the model is working fine.
- ❖ The precision of training set for labour (1) is 0.89 while that of testing is 0.85 too.

1.6 Model Tuning, Bagging (Random Forest should be applied for Bagging), and Boosting.

- **Model Tuning (Logistic regression) :**

- ❖ Model is tuned using gridsearch cv and the hyper-parameters chosen in this model building is 'penalty': 'l2', 'solver': 'newton-cg'
- ❖ The performance of this model is almost same as that of basic model.

- **Model Tuning (KNN) :**

- ❖ The hyper-parameters chosen in this model is 'n_neighbors': 17, 'weights': 'distance'
- ❖ The performance of this model is slightly improved as recall for labour in the testing dataset now is 0.91

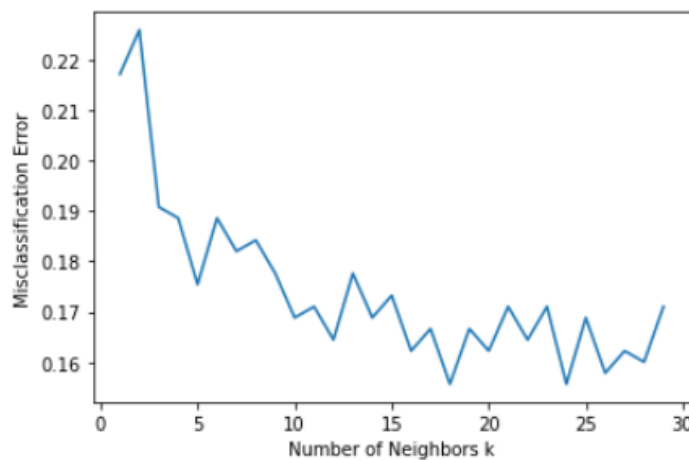


Fig 30. MCE

- ❖ From the above figure we can conclude that the optimal value of k where misclassification error is least is 17 or 24 so here k=17 is chosen for model building.

- **Adaboosting model performance :**
 - ❖ Training set accuracy : 0.842
 - ❖ Auc_roc score : 0.898
 - ❖ Testing set accuracy : 0.820
 - ❖ Auc_roc score testing : 0.878
- **Gradient boosting model performance :**
 - ❖ Training set accuracy : 0.892
 - ❖ Auc_roc score : 0.951
 - ❖ Testing set accuracy : 0.835
 - ❖ Auc_roc score testing : 0.899
- **Bagging :**
 - ❖ Bagging model is built with base estimator as random forest classifier and random state as 1 and n_estimators as 50.
 - ❖ Training set accuracy : 0.965
 - ❖ Auc_roc score : 0.997
 - ❖ Testing set accuracy : 0.828
 - ❖ Auc_roc score testing : 0.895

1.7 Performance Metrics: Check the performance of Predictions on Train and Test sets using Accuracy, Confusion Matrix, Plot ROC curve and get ROC_AUC score for each model. Final Model: Compare the models and write inference which model is best/optimized.

- For logistic regression base model is chosen as both the models are performing exactly same so either of the one can be chosen.
- For LDA, Naïve Bayes also the basic model is chosen and the performance metrics are already depicted above.
- For KNN, the tuned model is adopted.

	precision	recall	f1-score	support
0	1.00	1.00	1.00	307
1	1.00	1.00	1.00	754
accuracy			1.00	1061
macro avg	1.00	1.00	1.00	1061
weighted avg	1.00	1.00	1.00	1061

Table 14. classification report (tuned knn-training)

	precision	recall	f1-score	support
0	0.79	0.65	0.72	153
1	0.84	0.91	0.88	303
accuracy			0.83	456
macro avg	0.82	0.78	0.80	456
weighted avg	0.82	0.83	0.82	456

Table 15. classification report (tuned knn-testing)

- ❖ From the above classification report we can conclude that for class 0 the problem of overfitting is there.

- **Adaboost performance metrics :**

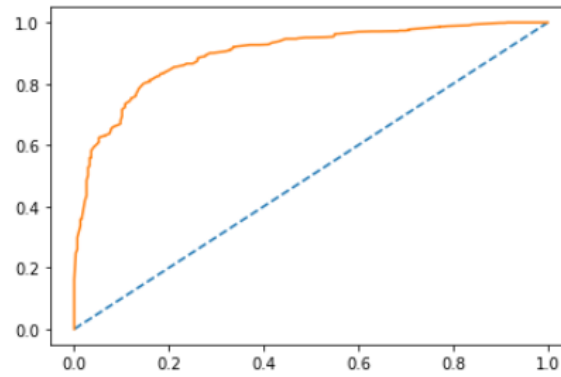


Fig 31. Auc_roc curve (adaboost training)

	precision	recall	f1-score	support
0	0.76	0.67	0.71	307
1	0.87	0.91	0.89	754
accuracy			0.84	1061
macro avg	0.81	0.79	0.80	1061
weighted avg	0.84	0.84	0.84	1061

Table 16. classification report (adaboost training)

- ❖ The recall for class 1 (labour) is 0.91 while that of class 0 (conservative) is 0.67
- ❖ The precision for class 1 is 0.87 while that of class 0 is 0.76

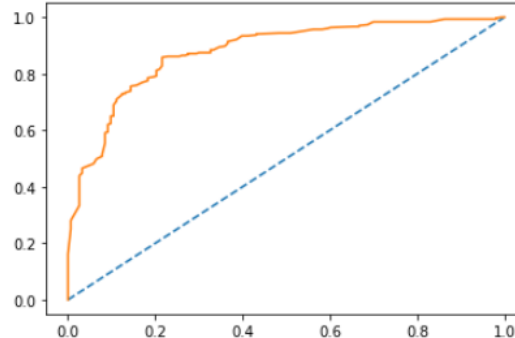


Fig 32. Auc_roc (adaboost testing)

	precision	recall	f1-score	support
0	0.74	0.72	0.73	153
1	0.86	0.87	0.87	303
accuracy			0.82	456
macro avg	0.80	0.80	0.80	456
weighted avg	0.82	0.82	0.82	456

Table 17. classification report (adaboost testing)

- ❖ On comparing the classification reports of training and testing set we can conclude that the model is working fine there is no under-fitting or over-fitting as the important metric here is precision and recall.

- Gradient boosting performance metrics

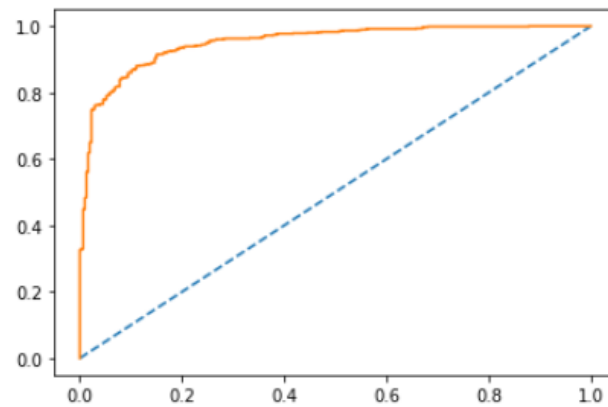


Fig 33. Auc_roc curve (gradient boostin training)

	precision	recall	f1-score	support
0	0.84	0.78	0.81	307
1	0.91	0.94	0.93	754
accuracy			0.89	1061
macro avg	0.88	0.86	0.87	1061
weighted avg	0.89	0.89	0.89	1061

Table 18. classification report (gradient boosting training)

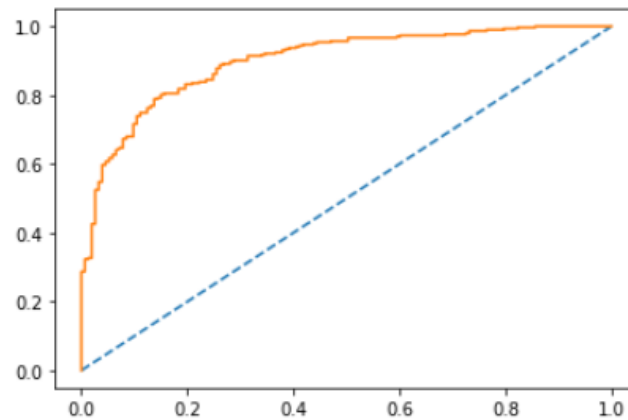


Fig 34. Auc_roc (gradient boosting-testing)

	precision	recall	f1-score	support
0	0.80	0.69	0.74	153
1	0.85	0.91	0.88	303
accuracy			0.84	456
macro avg	0.82	0.80	0.81	456
weighted avg	0.83	0.84	0.83	456

Table 19. classification report (gradient boosting – testing)

❖ By looking at the recall and precision score of both training and testing set we can conclude that the model is working fine.

- **Bagging performance metrics :**

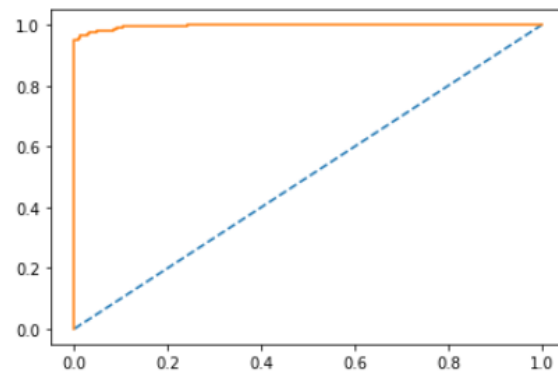


Fig 35. Auc_roc curve (bagging training)

	precision	recall	f1-score	support
0	0.98	0.90	0.94	307
1	0.96	0.99	0.98	754
accuracy			0.97	1061
macro avg	0.97	0.95	0.96	1061
weighted avg	0.97	0.97	0.96	1061

Table 20. classification report (bagging training)

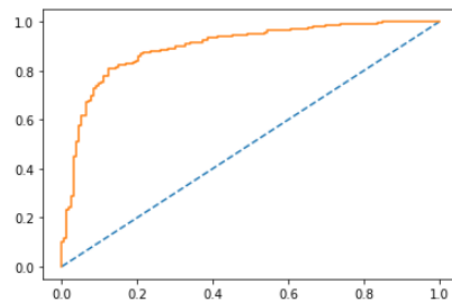


Fig 36. Auc_roc (bagging testing)

	precision	recall	f1-score	support
0	0.79	0.67	0.73	153
1	0.85	0.91	0.88	303
accuracy			0.83	456
macro avg	0.82	0.79	0.80	456
weighted avg	0.83	0.83	0.83	456

Table 21. classification report (bagging testing)

- ❖ From the above classification reports we can conclude that the model is slightly over-fitted as the precision of (1) in training set is 0.96 while in testing set is 0.85.

- **Comparison of different models on testing set for class 1 i.e. Labour.**

	Logistic Regression	LDA	Naïve Bayes	KNN (basic model)	Bagging	Adaboost	Gradient boost
Precision	0.86	0.86	0.87	0.85	0.85	0.86	0.85
Recall	0.88	0.89	0.87	0.89	0.91	0.87	0.91

Inferences :

- Here, precision and recall are the most important metrics
- The model performance of Logistic regression is not improved after applying grid search cv thus, it is a good model already.
- We can choose logistic regression, LDA, boosting or naïve bayes for model building process as they have shown consistent performance in training and testing set.

1.8 Based on these predictions, what are the insights?

- Here recall and precision is the most important metric.
- As per the comparison logistic regression, LDA, boosting or naïve bayes can be used for further model building.
- As per the multivariate analysis there exists no correlation among the variables.
- Many old age people prefer conservative party compared to labour party.
- More ratings in terms of national and household economic conditions are given to labour compared to conservative party.
- The disparity in performance of conservative party in training and testing set can be due to slight class imbalance also.

Problem 2:

In this particular project, we are going to work on the inaugural corpora from the nltk in Python. We will be looking at the following speeches of the Presidents of the United States of America:

President Franklin D. Roosevelt in 1941

President John F. Kennedy in 1961

President Richard Nixon in 1973

2.1 Find the number of characters, words, and sentences for the mentioned documents.

The number of sentences = 5217

The number of words = 152901

The number of characters = 667512

2.2 Remove all the stopwords from all three speeches.

The stopwords that are part of the given speech are :

['i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', "you're", "you've", "you'll", "you'd", 'your', 'yours', 'yo
urself', 'yourselves', 'he', 'him', 'his', 'himself', 'she', "she's", 'her', 'hers', 'herself', 'it', "it's", 'its', 'itself', 'they', 'the
m', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'whom', 'this', 'that', "that'll", 'these', 'those', 'am', 'is', 'are', '
was', 'were', 'be', 'been', 'being', 'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doing', 'a', 'an', 'the', 'and', 'but', 'if', '
or', 'because', 'as', 'until', 'while', 'of', 'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into', 'through', 'during', 'be
fore', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'out', 'on', 'off', 'over', 'under', 'again', 'further', 'then', 'on
ce', 'here', 'there', 'when', 'where', 'why', 'how', 'all', 'any', 'both', 'each', 'few', 'more', 'most', 'other', 'some', 'such', '
no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'too', 'very', 's', 't', 'can', 'will', 'just', 'don', "don't", 'should', "should

've', 'now', 'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', 'aren't', 'couldn', 'couldn't', 'didn', 'didn't', 'doesn', 'doesn't', 'hadn', 'hadn't', 'hasn', 'hasn't', 'haven', 'haven't', 'isn', 'isn't', 'ma', 'mightn', 'mightn't', 'mustn', 'mustn't', 'needn', 'needn't', 'shan', 'shan't', 'shouldn', 'shouldn't', 'wasn', 'wasn't', 'weren', 'weren't', 'won', 'won't', 'wouldn', 'wouldn't', '!', '"', '#', '\$', '%', '&', "'", '(', ')', '*', '+', ',', '-', '.', '/', ':', ';', '<', '=', '>', '?', '@', '[', '\\', ']', '^', '_', '`', '{', '|', '}', '~']

2.3 Which word occurs the most number of times in his inaugural address for each president? Mention the top three words. (after removing the stopwords)

Kennedy speech :

('', 85), ('the', 83), ('of', 65)

Nixon speech :

('', 96), ('the', 80), ('.', 68)

Roosevelt speech :

('the', 104), ('of', 81), ('.', 77)

